Reflective Journal: Module 06 Lab - Regression and Classification

For Module 06, we focused on regression and classification. This lab helped me understand the difference between the two a lot better because we actually used both of them in code instead of just reading about them. Before doing the lab, I understood the basic definition, but working through the Titanic dataset made it more clear. Regression is used when we are trying to predict a number, and classification is used when we are trying to predict a category.

In the Linear Regression part, we used the model to predict the passenger fare. This made sense as a regression problem because fare is a number and can have many different values. The model was not trying to say yes or no. It was trying to estimate an amount. Looking at the mean squared error helped me understand that regression is not judged the same way as classification. With fare prediction, the model can be a little off, very off, or close enough, so we need a metric that measures the size of the error.

The Logistic Regression part was different because we used it to predict whether a passenger survived or not. Even though the name has "regression" in it, this part helped me remember that Logistic Regression is really used for classification. The output is more like a category, such as survived or did not survive. Accuracy made more sense here because the prediction is either correct or incorrect.

One thing I liked about this lab was looking at the coefficients. They helped me see that the model is not just giving random predictions. The coefficients give an idea of how each feature is affecting the result. I still do not feel like I understand every detail perfectly, but I understand the general idea better now. A positive coefficient means the feature pushes the prediction upward, while a negative coefficient pushes it downward.

The hardest part for me was keeping the two sections separate in my head. Since both sections used the Titanic dataset, it was easy to mix up what we were predicting. I had to remind myself that predicting `Fare` was regression, while predicting `Survived` was classification. Once I slowed down and focused on the target variable, the difference made more sense.

This lab also reminded me that data preparation still matters. Even though this module was about models, the data still had to be cleaned and converted into a format the model could use. For example, the `Sex` column had to be turned into numbers before Logistic Regression could use it. That connected back to what we learned in the earlier data preparation module.

Overall, this lab helped me understand supervised learning better. I learned that the type of target variable decides which kind of model and evaluation metric should be used. If the target is a number, regression makes sense. If the target is a category, classification makes sense. This was one of the clearer labs for me because I could see the difference directly in the code and outputs.